

Assignment: Azure Cloud Fundamentals and Data Pipeline Implementation using ADF

Objective: To understand Azure cloud concepts and build a complete data pipeline using Storage Account and Azure Data Factory.

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Task 1: Explore Azure Portal & Create a Resource Group

A Resource Group in Microsoft Azure is a logical container used to group related Azure resources such as storage accounts, data factories, and virtual machines. It allows resources belonging to a single project or solution to be managed, monitored, and billed together. In this task, a resource group named "rg-dataengineering" was created to organize all resources used throughout this project.

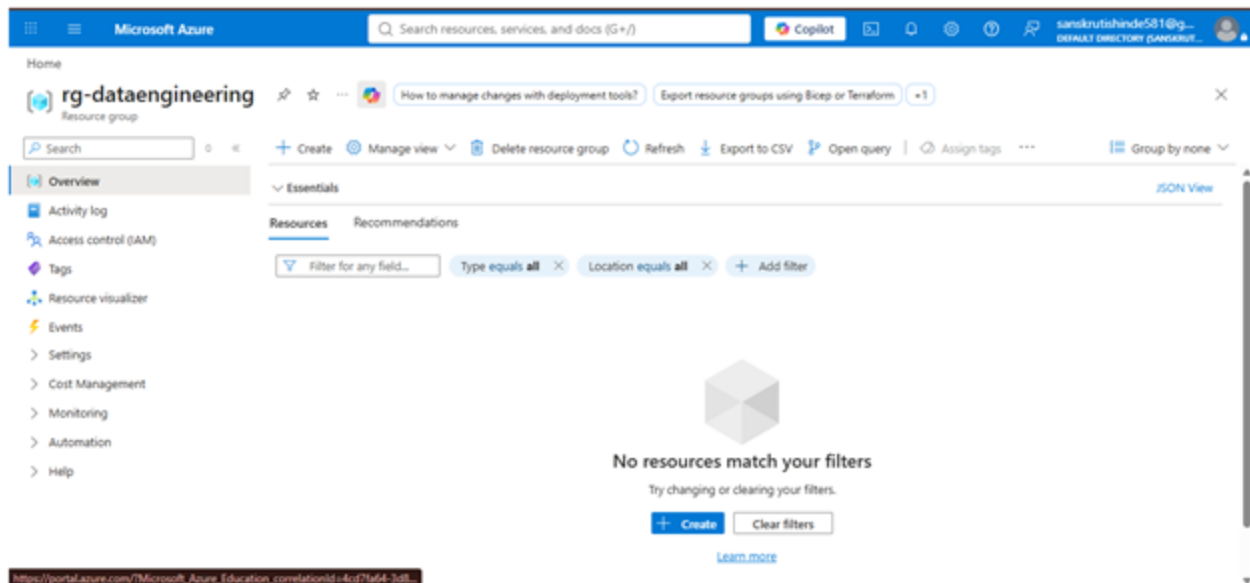


Figure 1: Resource group "rg-dataengineering" created in the Azure Portal.

Task 2: Storage Setup

A Storage Account is a Microsoft Azure service that provides scalable, durable cloud storage for data such as files, blobs, and tables. Within a storage account, a Blob Container acts as a folder-like structure used to organize and store unstructured data such as CSV files. In this task, a storage account was created, a container was added within it, and a sample CSV file was uploaded to serve as the data source for the pipeline.

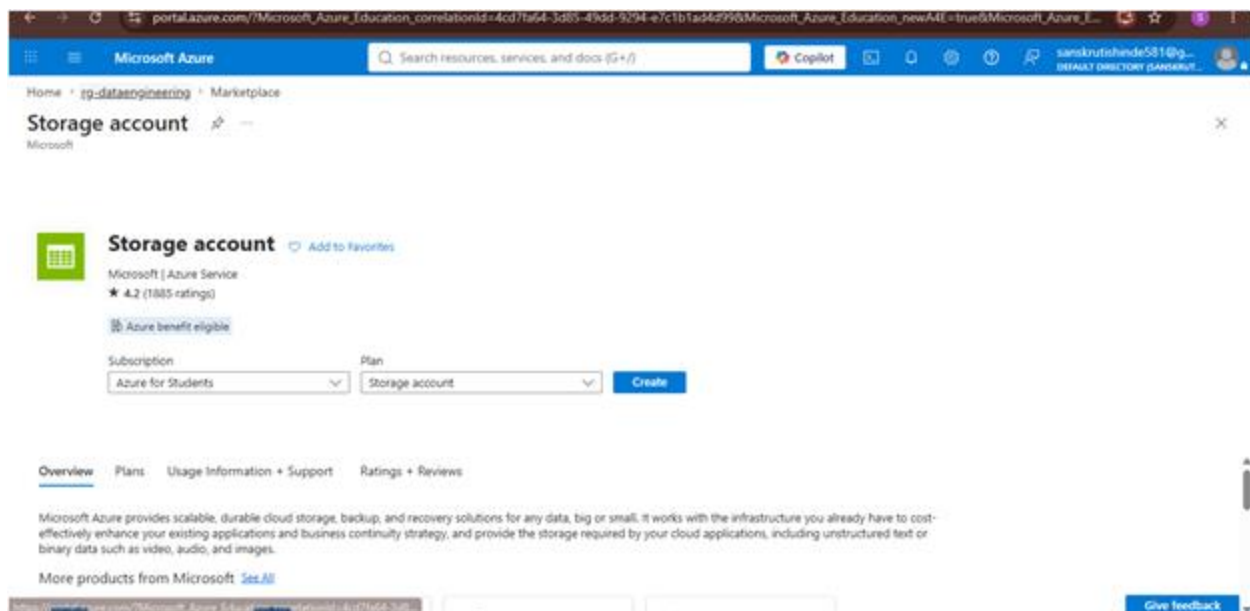


Figure 2: Storage account creation option selected from the Azure Marketplace.

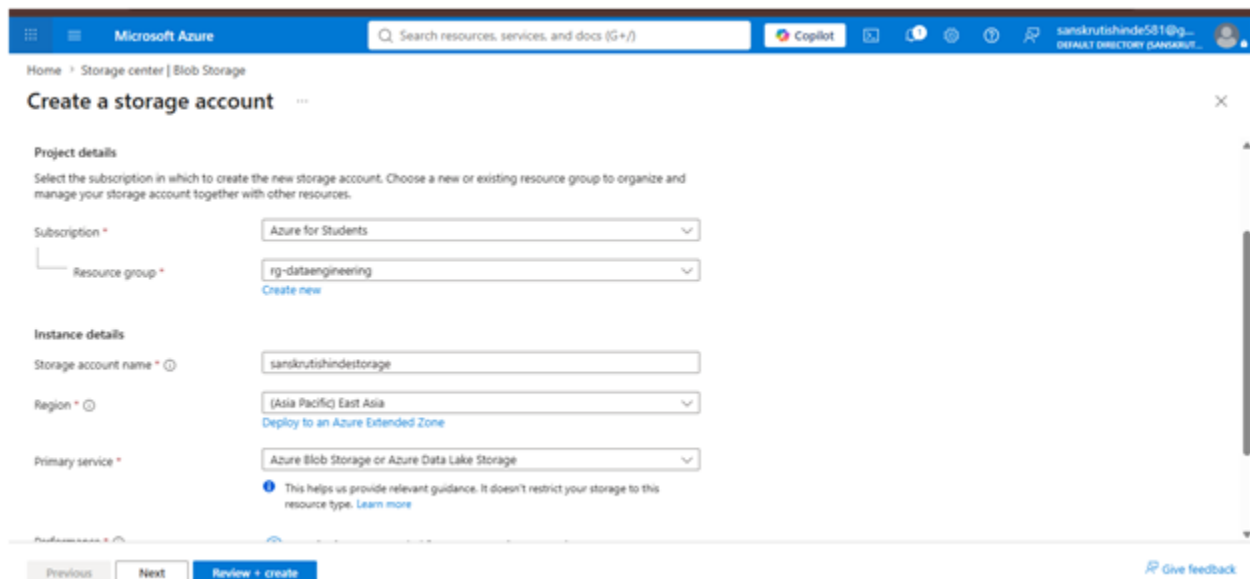


Figure 3: Storage account configuration with subscription, resource group, and region details.

Home > rg-dataengineering > Marketplace > Storage account

Create a storage account

Project details
Select the subscription in which to create the new storage account. Choose a new or existing resource group to organize and manage your storage account together with other resources.

Subscription * ▼ Azure for Students

Resource group * ▼ rg-dataengineering
[Create new](#)

Instance details

Storage account name * ⓘ

Region * ⓘ ▼ (Asia Pacific) Central India
[Deploy to an Azure Extended Zone](#)

Primary service * ▼ Azure Blob Storage or Azure Data Lake Storage

! This helps us provide relevant guidance. It doesn't restrict your storage to this resource type. [Learn more](#)

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Figure 4: Storage account creation reattempted with the region set to Central India.

Home > Storage center | Blob Storage > sanskrutishindestorage

sanskrutishindestorage | Containers

Storage account

× + Add container ↑ Upload ↻ Refresh 🗑️ Delete 🔒 Change access level 🔄 Restore containers ⚙️ Edit columns

Only show active containers ▼

Showing all 1 items

☐	Name	Last modified	Anonymous access level	Lease state	Storage connector
☐	con	7/12/2026, 9:58:22 AM	Private	Available	-

Add or remove favorites by pressing CTRL+SHIFT+F

Figure 5: Blob container view showing the existing container within the storage account.

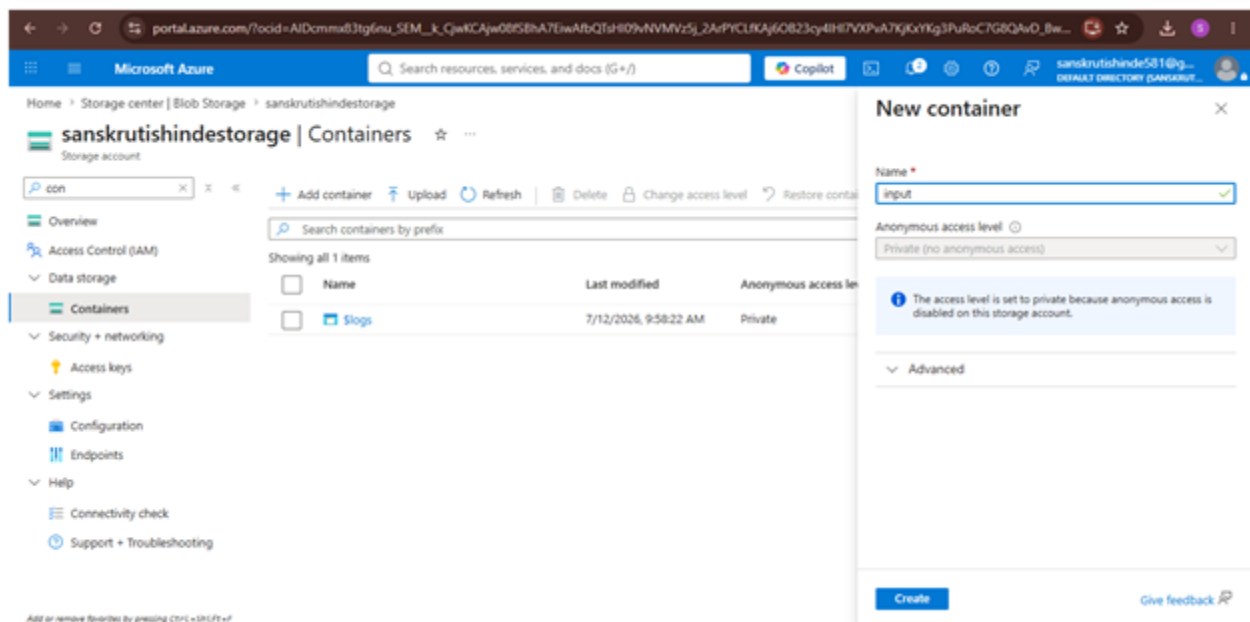


Figure 6: Creation of a new container named "input" for storing source data.

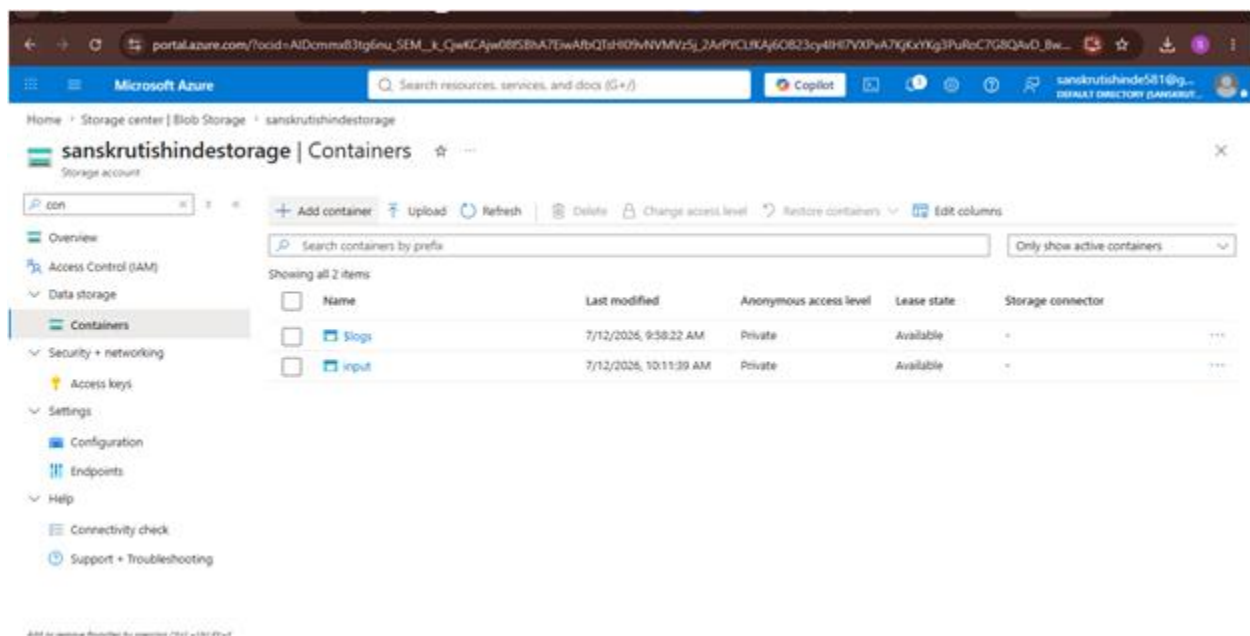


Figure 7: Containers list showing both containers present in the storage account.

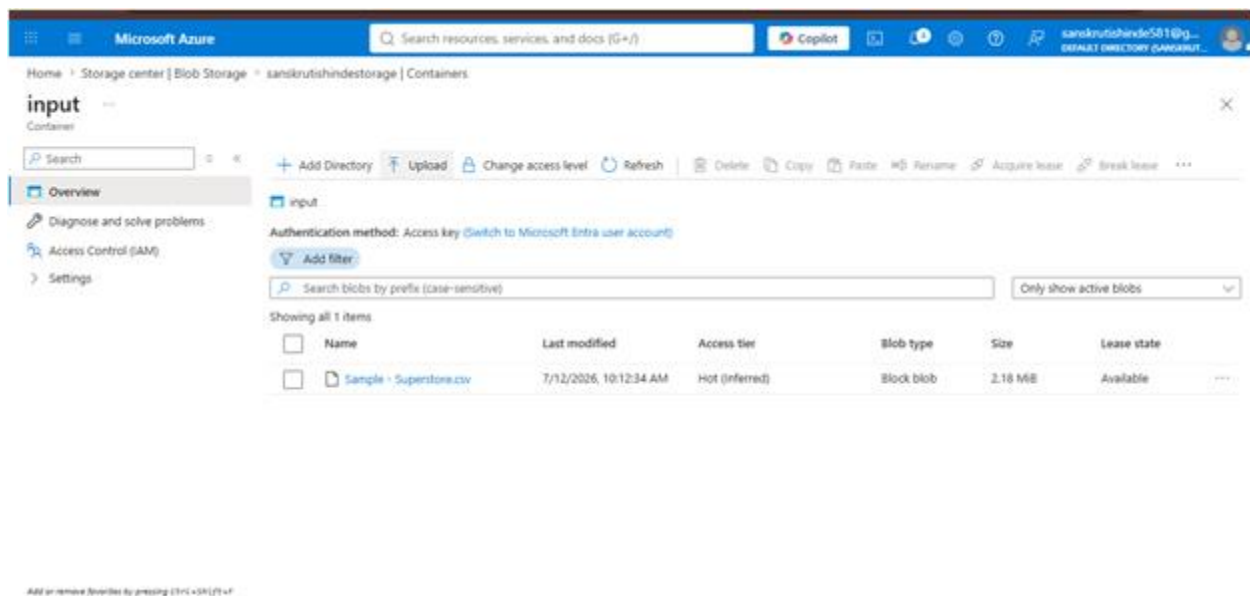


Figure 8: Sample CSV file ("Sample - Superstore.csv") uploaded successfully into the input container.

Task 3: ADF Basics

Azure Data Factory (ADF) is a cloud-based data integration service that allows creation of data-driven workflows for orchestrating and automating data movement and transformation. A Linked Service defines the connection information required to connect ADF to an external data store, such as a storage account. Datasets represent the structure of data within the linked data stores, and the Get Metadata activity is used to retrieve metadata (such as file existence or properties) about a specified dataset before further processing.

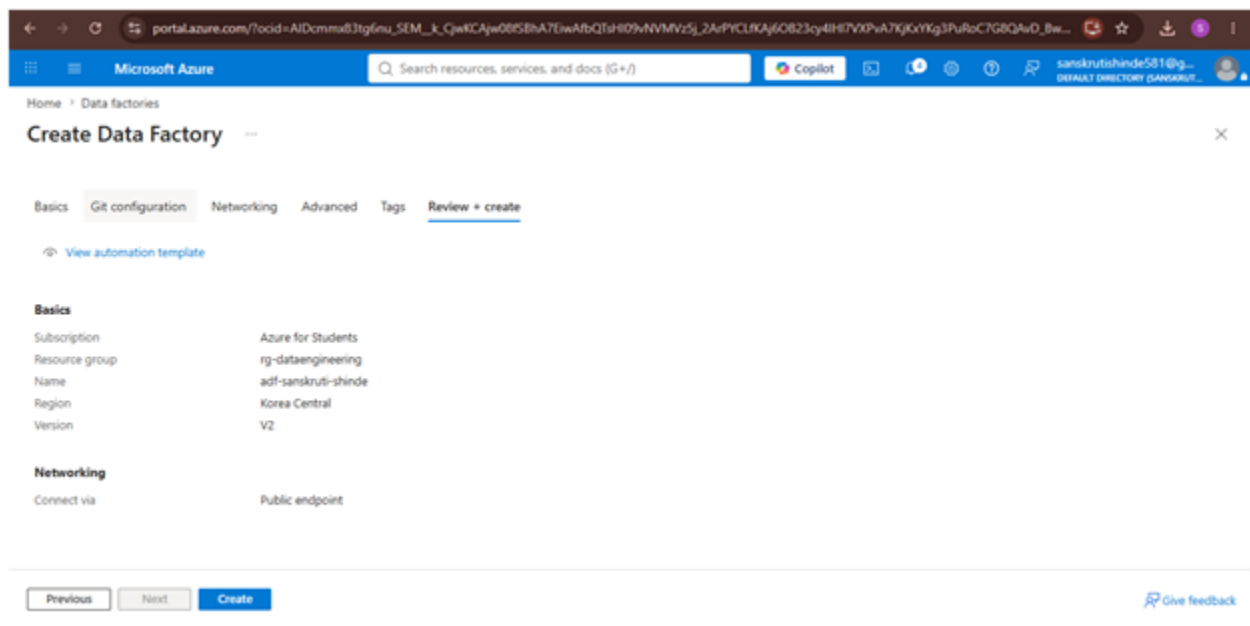


Figure 9: Review and creation of the Azure Data Factory instance "adf-sanskriti-shinde".

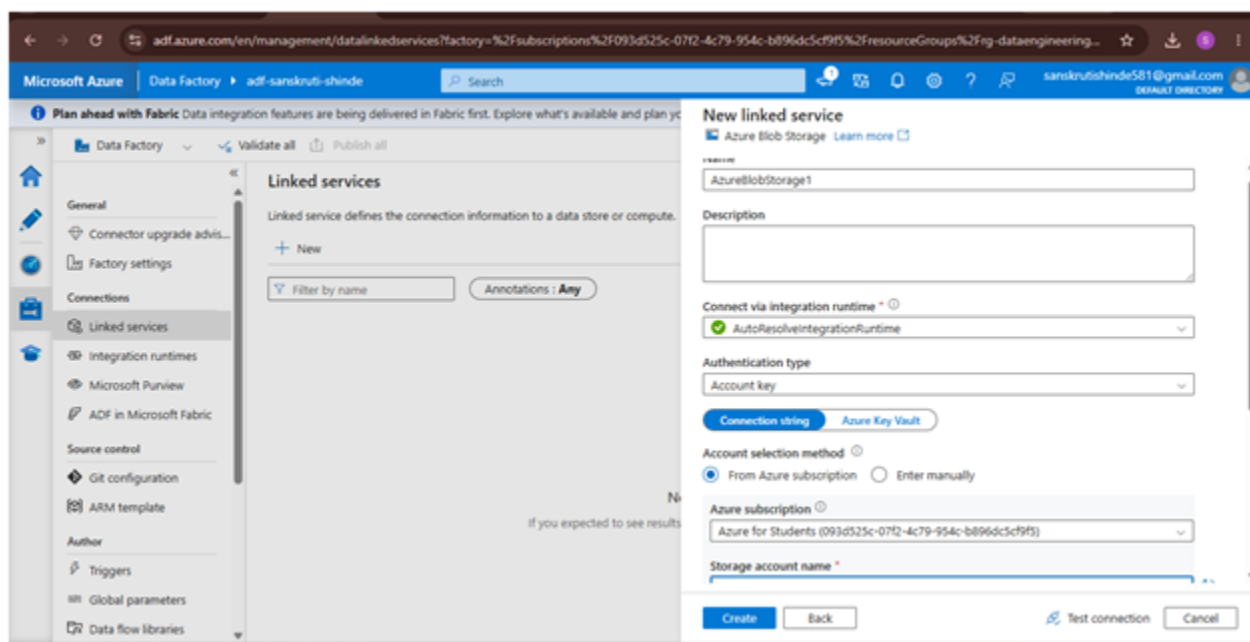


Figure 10: Configuration of a new Linked Service ("AzureBlobStorage1") connecting ADF to the storage account.

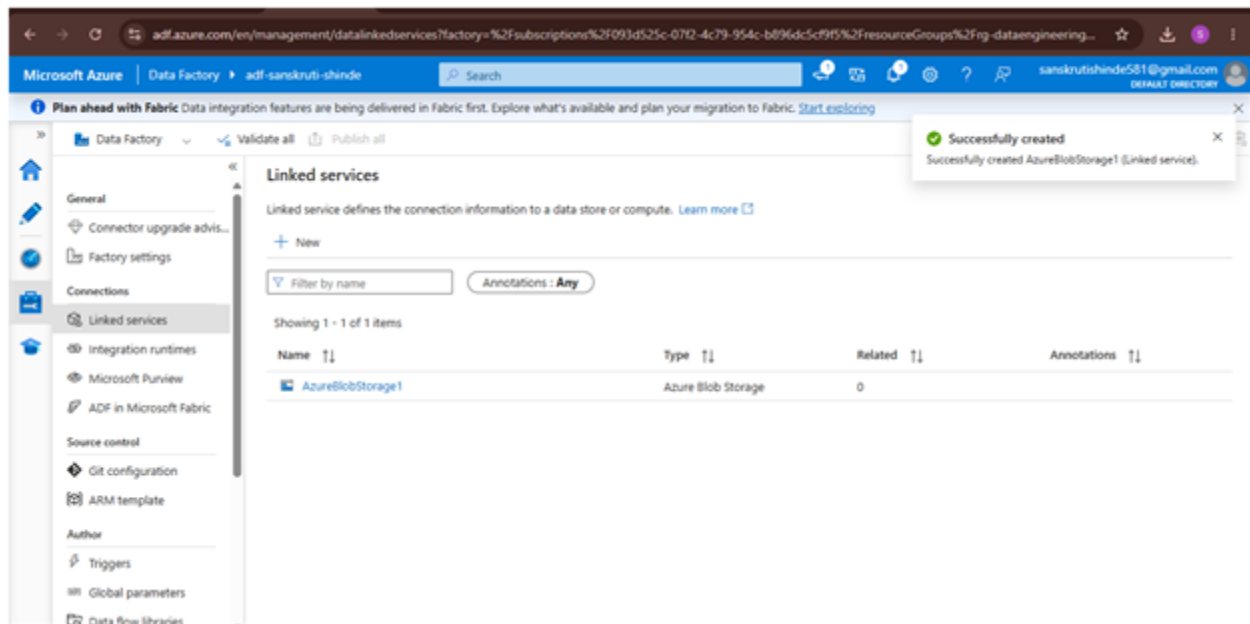


Figure 11: Linked Service "AzureBlobStorage1" successfully created.

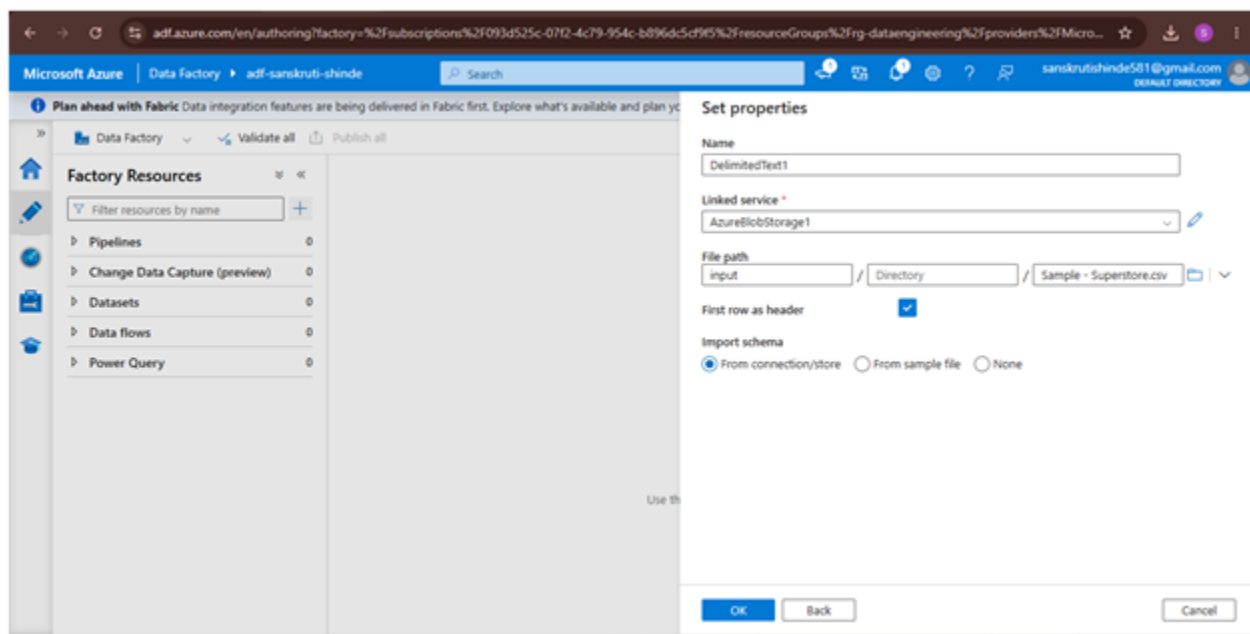


Figure 12: Source dataset ("DelimitedText1") configured to point to the CSV file in the input container.

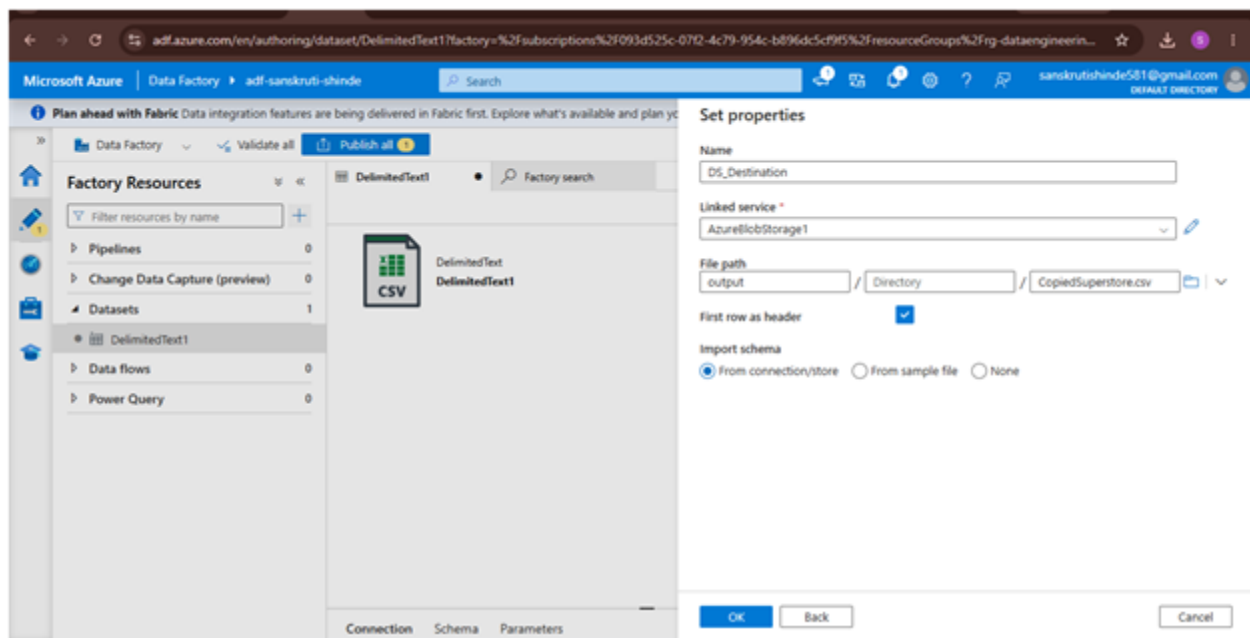


Figure 13: Destination dataset ("DS_Destination") configured to write output to the output container.

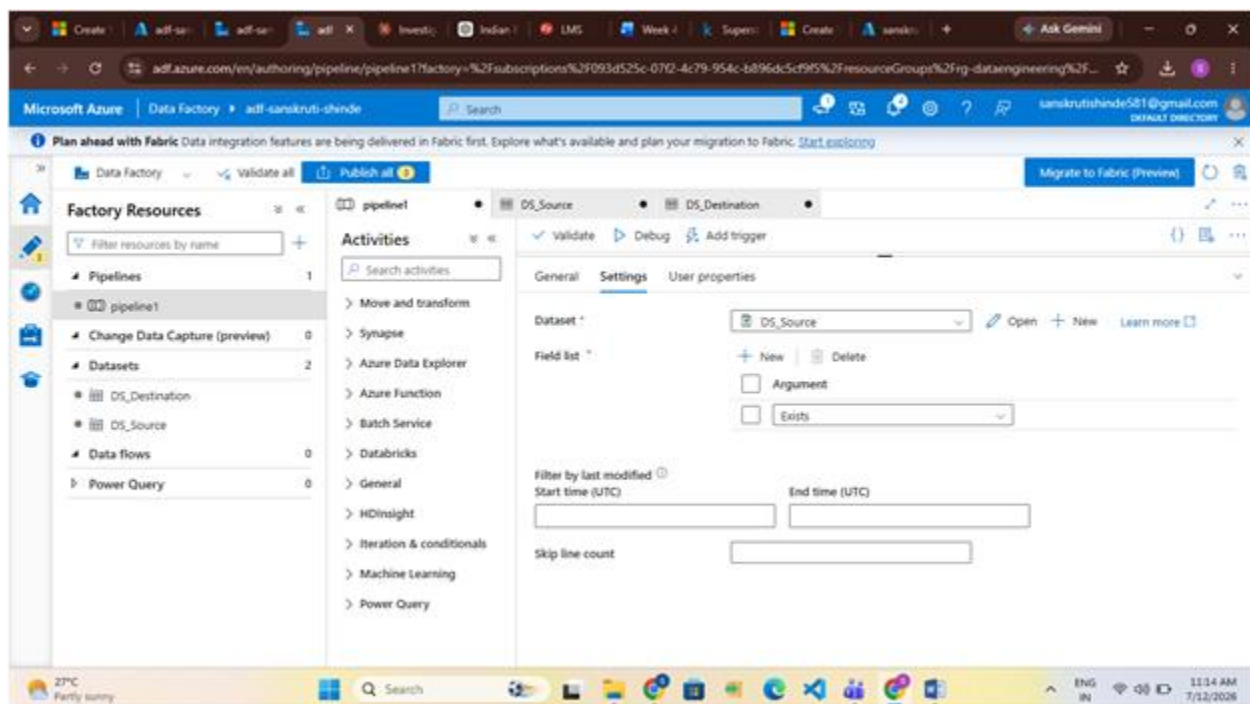


Figure 14: Pipeline settings for the Get Metadata activity used to validate the source dataset.

Task 4: Pipeline Development

The Copy Data activity in Azure Data Factory is used to copy data from a source dataset to a destination dataset, optionally applying schema mapping along the way. In this task, a pipeline was designed by connecting a Get Metadata activity with a Copy Data activity, ensuring that metadata validation occurs before the actual data copy operation. The pipeline was then validated and published to the live ADF environment.

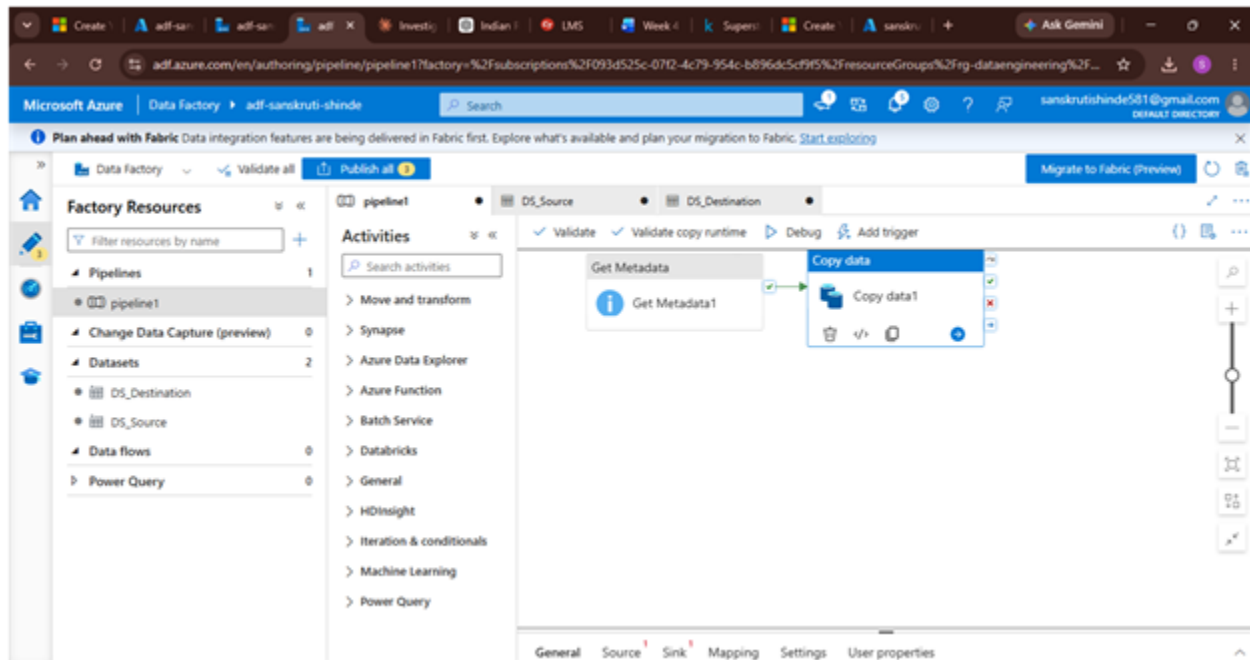


Figure 15: Pipeline canvas showing the Get Metadata and Copy Data activities connected together.

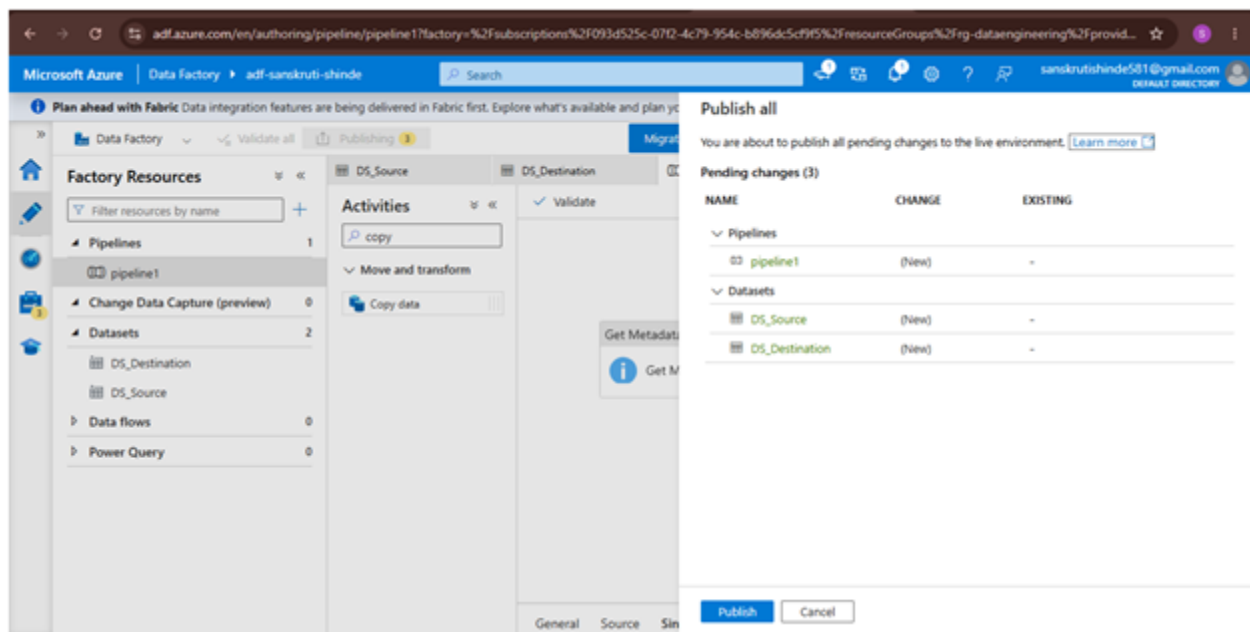


Figure 16: Publishing pending changes, including the pipeline and its associated datasets, to the live environment.

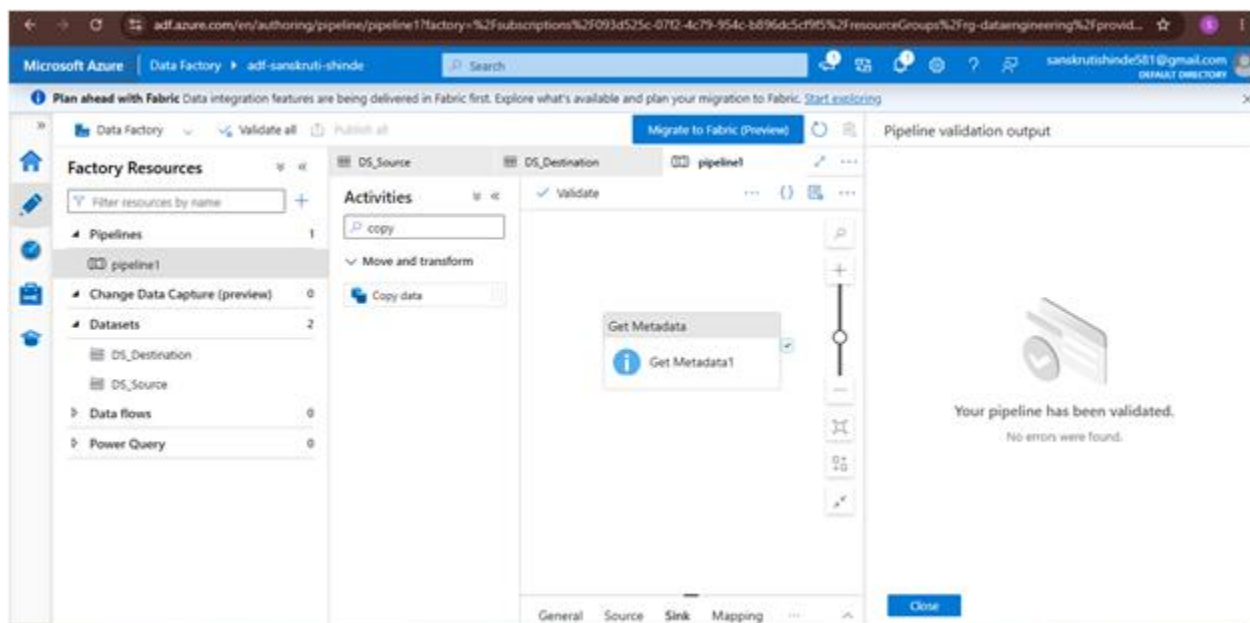


Figure 17: Pipeline validation output confirming that no errors were found.

Task 5: Pipeline Execution

Once a pipeline has been designed and validated, it can be executed using the Debug option in Azure Data Factory, which allows the pipeline to be tested without publishing it as a trigger-based run. Running the pipeline in debug mode helps confirm that all activities execute successfully and that data flows correctly from the source to the destination.

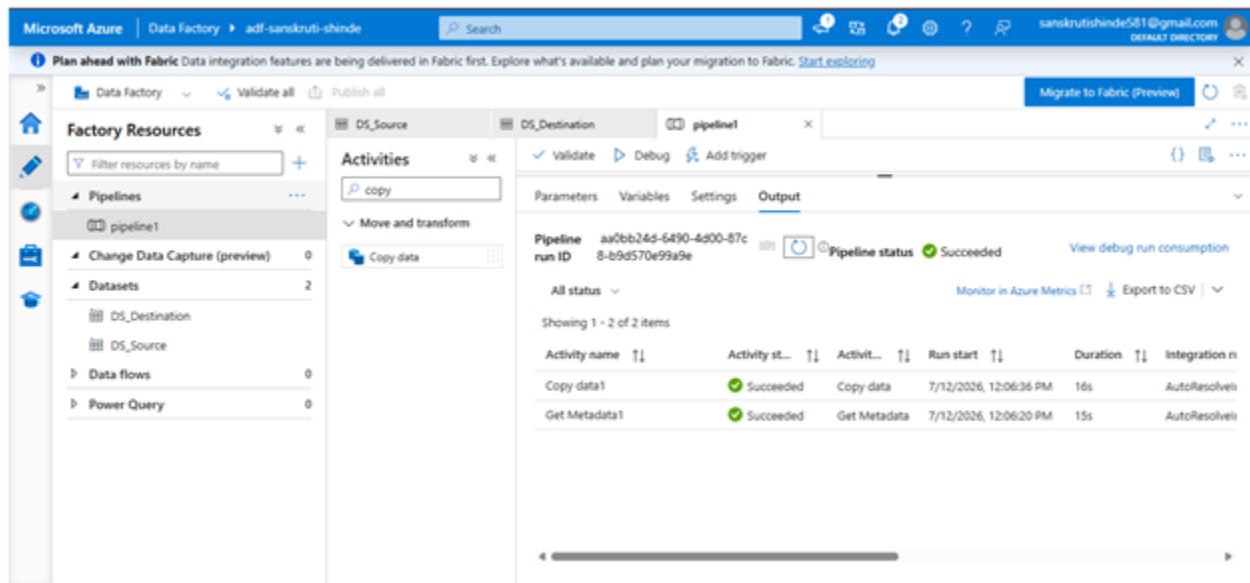


Figure 18: Pipeline run output showing both the "Get Metadata" and "Copy Data" activities with a status of Succeeded.

Task 6: IAM Roles

Identity and Access Management (IAM) in Azure allows fine-grained control over who or what can access a resource, and what actions they are permitted to perform. The Reader role grants read-only access to resources, while the Storage Blob Data Contributor role grants read, write, and delete permissions on blob data specifically. Rather than granting these permissions to a personal user account, access should be granted to Azure Data Factory's Managed Identity. A Managed Identity is an automatically managed identity tied directly to the ADF instance, which allows it to securely authenticate to the storage account without embedding or exposing any credentials, following the principle of least privilege and improving overall security.

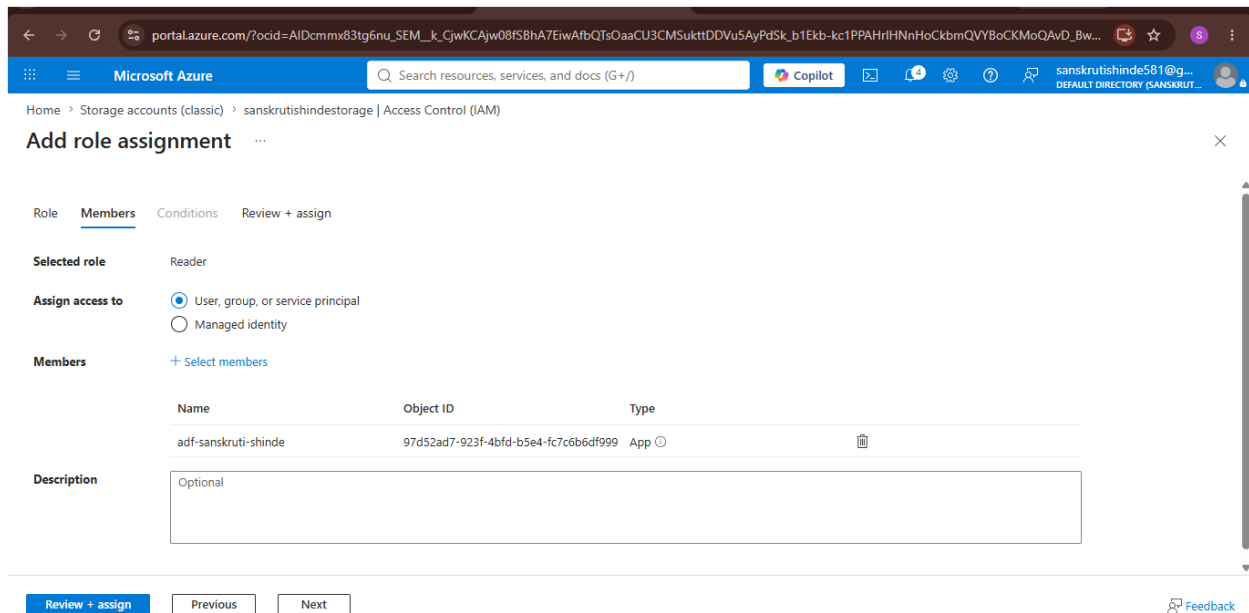


Figure 19: Role assignment granting the "Reader" role to the ADF managed identity ("adf-sanskriti-shinde").

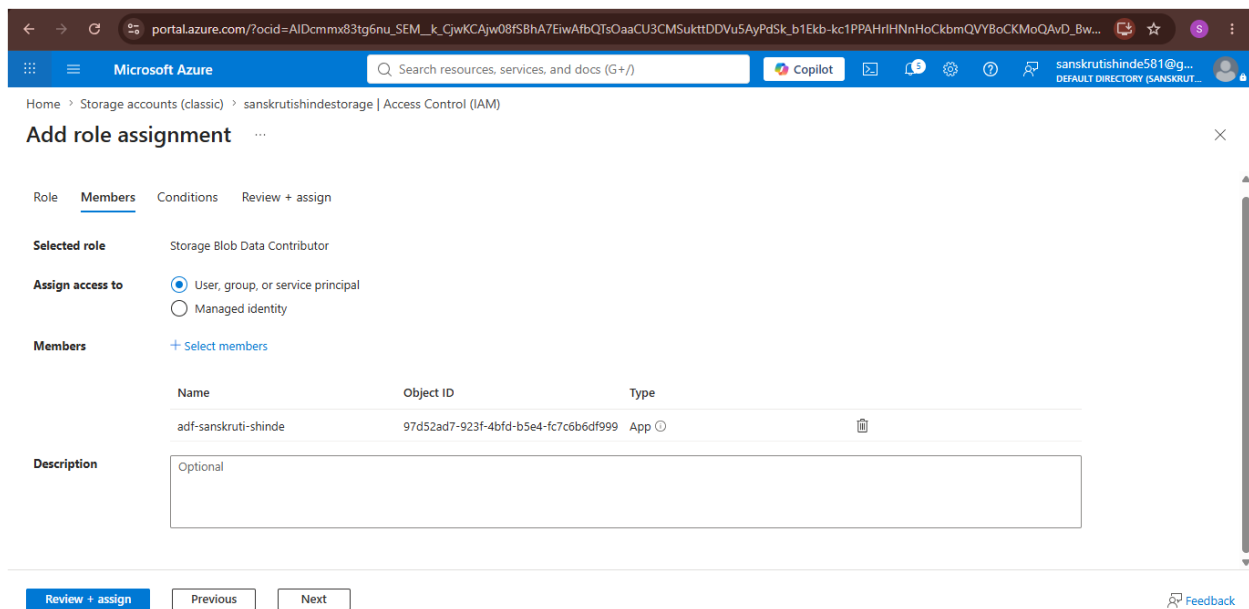


Figure 20: Role assignment granting the "Storage Blob Data Contributor" role to the ADF managed identity.

• Mini Project: End-to-End Pipeline

Problem Statement:

To build a complete, end-to-end data pipeline that reads a CSV file from Azure Blob Storage and processes it using Azure Data Factory, ultimately copying the processed data to a separate destination container.

Summary:

- Source: "Sample - Superstore.csv" file stored in the "input" container of the storage account "sanskrutishindestorage".
- Linked Service, Dataset, and Pipeline Used: The linked service "AzureBlobStorage1" connected ADF to the storage account; the source dataset "DelimitedText1" and destination dataset "DS_Destination" were built on top of this linked service; both were orchestrated within "pipeline1".
- Process: The pipeline first executed a Get Metadata activity to validate the existence of the source file, followed by a Copy Data activity that transferred the file contents to the destination container.
- Destination: The processed file was copied as "CopiedSuperstore.csv" into the "output" container.

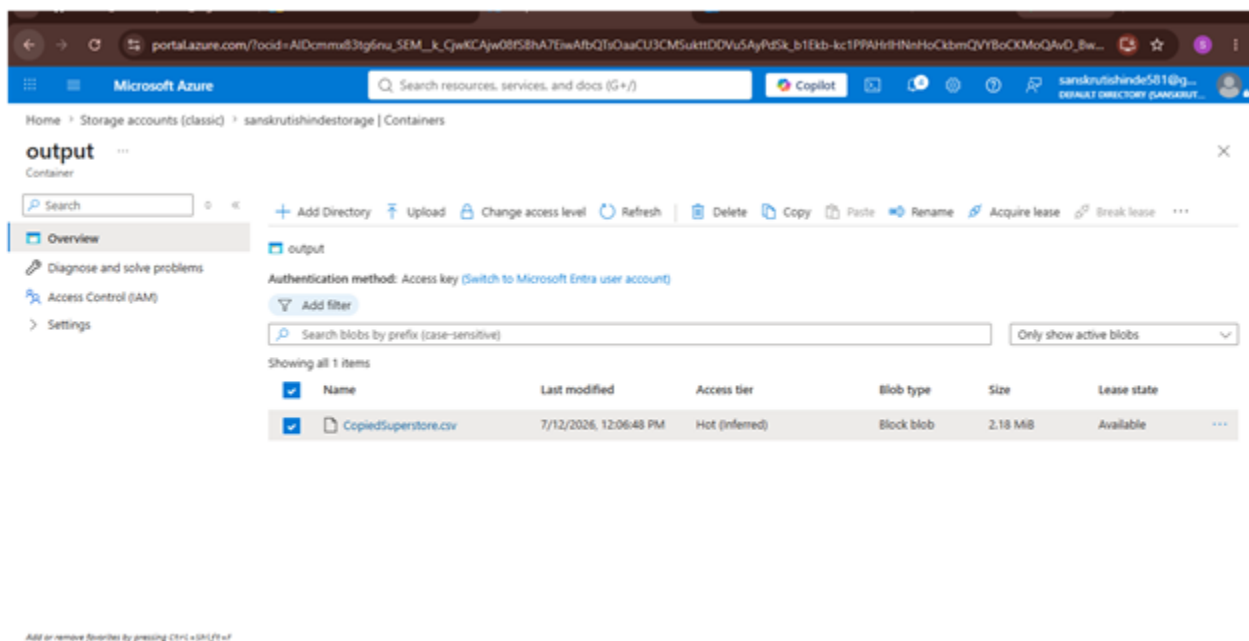


Figure 21: Output container showing the copied file "CopiedSuperstore.csv" after successful pipeline execution.

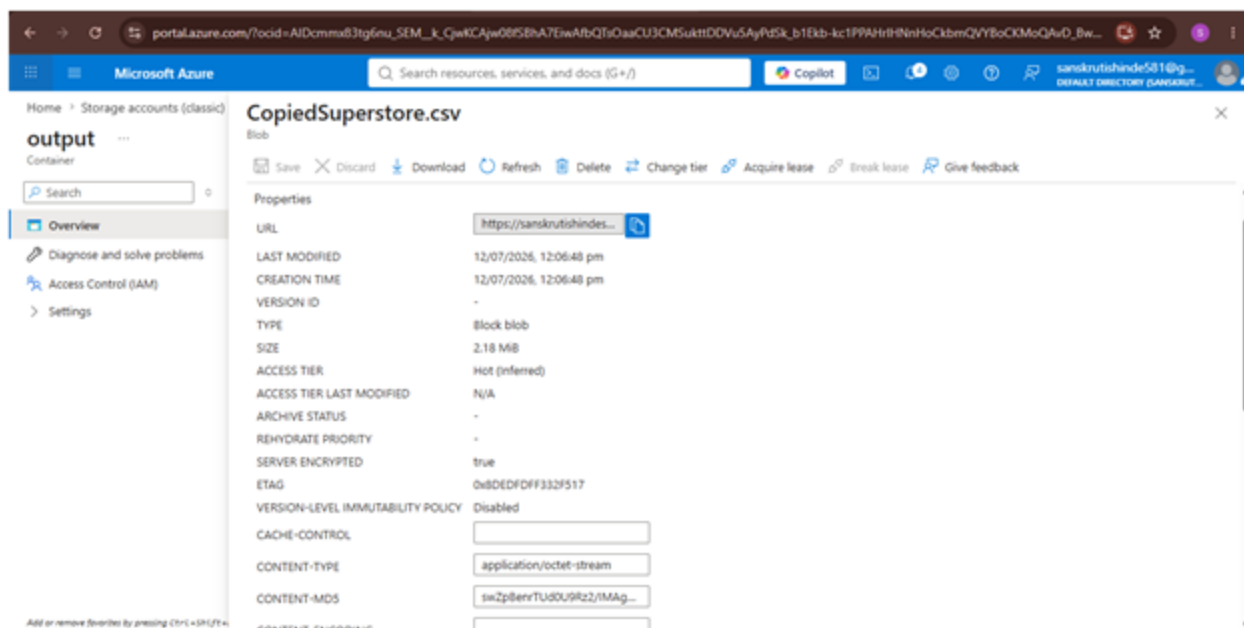


Figure 22: Properties of the "CopiedSuperstore.csv" blob confirming successful upload to the destination container.

Expected Outputs Confirmed:

- The pipeline executed successfully with a "Succeeded" status.
- Data was successfully copied from the source container to the destination container.
- Metadata of the source file was validated prior to the copy operation.

Conclusion

This assignment provided hands-on experience with core Azure cloud concepts, beginning with the creation of a resource group and progressing through the setup of a storage account and blob containers. It further covered the fundamentals of Azure Data Factory, including linked services, datasets, and the Get Metadata activity, followed by the design, validation, and execution of a complete data pipeline using the Copy Data activity. The mini project consolidated these individual tasks into a single end-to-end pipeline that successfully read a CSV file from Blob Storage and copied it to a destination container. Overall, the assignment strengthened understanding of cloud resource organization, secure access management through IAM and managed identities, and the orchestration of data movement using Azure Data Factory.